PumpGuard Ops pilot business case

Decision brief for the CFO and Board

KES 1.50M gated pilot

Recommendation: Approve a capped 12-month pilot for 24 pumps, subject to data-readiness and 90-day shadow-validation gates. The base case returns KES 2.58M in annual benefit, 72% first-year ROI, and a 7.0-month payback.

PROBLEM

Maintenance teams must compare vibration, temperature, pressure, current, and error readings before deciding which pump needs attention. The prototype workflow assumes this review takes 30 minutes per day for a 24-pump fleet. Manual review can delay investigation when several signals deteriorate together.

PROPOSED SOLUTION

PumpGuard Ops validates telemetry, calculates rolling sensor indicators, and ranks each pump by seven-day failure risk. Engineers receive a review queue, the model's main risk contributions, recent telemetry, and a downloadable action list. Humans retain every maintenance and shutdown decision.

BASE-CASE RETURN

72% ROI

7.0 months payback

Illustrative assumptions, not KPC actuals

FINANCIAL IMPACT

Calculation	Base assumption	Annual value
Avoided events	8 events x 25% prevention	**2 events**
Avoided disruption	2 events x KES 1.20M	**KES 2.40M**
Review time	28 min x 250 days x KES 1,500/hr	**KES 0.175M**
Gross annual benefit	KES 2.40M + KES 0.175M	**KES 2.575M**
First-year cost	Setup KES 0.90M + run cost KES 0.60M	**KES 1.500M**
Net benefit and ROI	(KES 2.575M - KES 1.500M) / KES 1.500M	**KES 1.075M / 72%**
Payback	KES 1.500M / KES 2.575M x 12	**7.0 months**

TOP RISKS AND CONTROLS

Model underperformance: Synthetic results may not transfer to real pumps. Run 90 days in shadow mode and require engineering sign-off before operational alerts.
Alert fatigue or poor data: Track sensor completeness and false alarms weekly. Escalate only critical cases until the team calibrates the threshold.

BOARD DECISION

Approve up to **KES 1.50M** for the 24-pump pilot. Release implementation funds only after the data audit. Continue after shadow validation only if recall reaches 80% and the false-positive rate stays below 5%. Scale only when verified savings exceed operating cost and engineers confirm safe use.